BUSFIN 8200 – Problem Set 2

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Question 1

Question 2

2 (a)

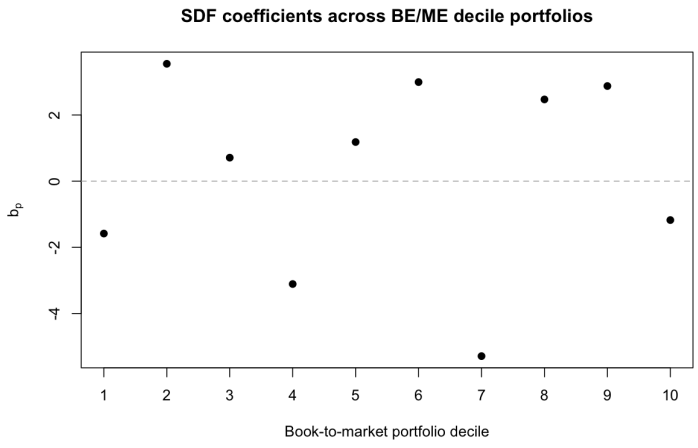
Using the normalization $\mathbb{E}[M^*] = 1$, we find the stochastic discount factor $M_t^* = a - b'xR_t$ that satisfies $\hat{\mathbb{E}}[M^* \cdot xR] = 0$. In particular, we have:

$$\begin{aligned} 0 &= \mathbb{E}[M^* \cdot xR] \\ &= \underbrace{\mathbb{E}[M^*]}_{=1} \cdot \mathbb{E}[xR] + \text{Cov}(M^*, xR) \\ &= \mathbb{E}[xR] - b'\text{Cov}(xR, xR) \\ &\implies \mathbb{E}[xR] = b'\Sigma \\ &\implies b = \Sigma^{-1}\mathbb{E}[xR] \end{aligned}$$

xR is a Tx10 vector of excess returns with T = 1,202 months in the sample from July 1926 to August 2026. Verifying that Σ is invertible, we find that:

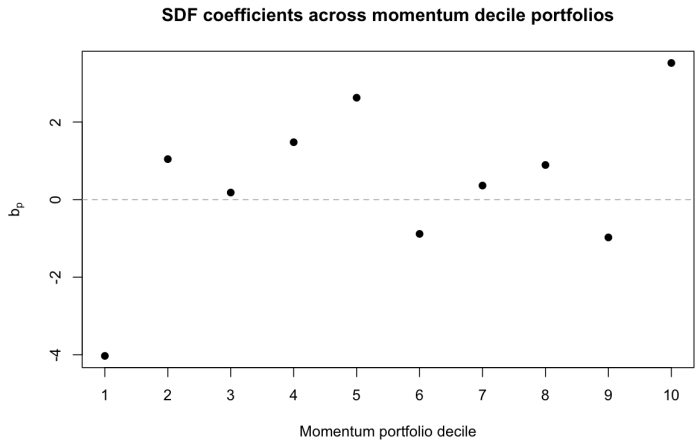
$$\begin{aligned} 1 &= \mathbb{E}[M^*] \\ &= \mathbb{E}[a - b'xR] \\ &= a - b'\mathbb{E}[xR] \\ &\implies a = 1 + b'\mathbb{E}[xR] \end{aligned}$$

Therefore, we are able to compute the stochastic discount factor $M_t^* = a - b'xR_t$ for each month in the sample and obtain the following scatterplot:



2 (b)

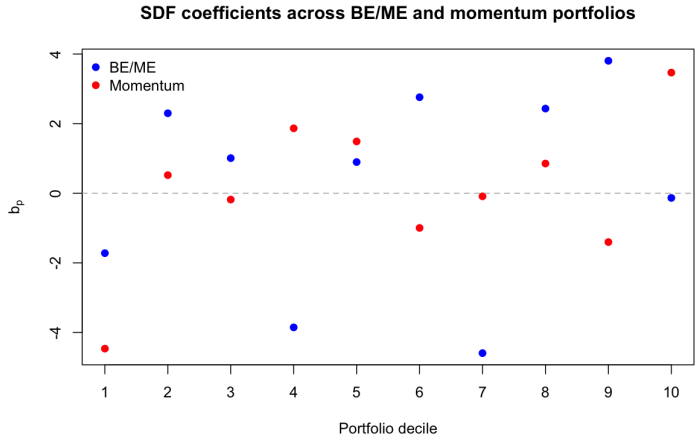
Following the same procedure as in part (a), we find the respective b coefficients of the stochastic discount factor that prices the excess returns of the 10 portfolios in the MOM dataset as follows:



Similar to part (a), we do not observe any clear pattern between the b coefficients and the decile portfolios' excess returns. The correlation between the recovered M_t^* is 0.3579, which suggests that the SDF that prices a particular portfolio does not necessarily price the other one as the decile portfolios are sorted using different signals.

2 (c)

Following the same procedure as in part (a), we find the respective b coefficients of the stochastic discount factor that prices the excess returns of the 20 portfolios in the SMB & MOM datasets as follows:



Using a common sample of 1,196 months from January 1927 to August 2026, we estimated the Sharpe ratios of xR_t^* as follows:

Table 1: Monthly Sharpe Ratios of the xR^* Portfolios

Portfolio set	Monthly Sharpe ratio
BE/ME (10 portfolios)	0.192030
Momentum (10 portfolios)	0.260270
Combined (20 portfolios)	0.306976

A key property of the SDF under no arbitrage is that xR_t^* is the maximum Sharpe ratio portfolio. Since we observed that the Sharpe ratio of the combined 20 portfolios is higher than the Sharpe ratios of the individual 10 portfolios, adding more non-redundant assets to the portfolio universe provides a more diversified portfolio with better risk-adjusted returns.

2 (d)

2 (e)

Question 3

3 (a)

3 (b)

3 (c)

3 (d)

3 (e)

Question 4

4 (a)

4 (b)

4 (c)

4 (d)

4 (e)